

TRADE BRILLIANTLY



thinkorswim® is now at Schwab.

Our award-winning thinkorswim trading platforms are loaded with powerful features that let you dive deeper into the market.

- Visualize your trades in a new light on thinkorswim desktop with robust charting and analysis tools, including 400+ technical studies.
- Uncover new opportunities with up-to-the-minute market news and insights.
- Choose a platform to fit your trading style—from streamlined to advanced.
- Available on desktop, web, and mobile to meet you where you are so you never miss a thing.

Built by the trading-obsessed, so you can trade brilliantly.



Investing involves risks, including loss of principal.
Schwab does not recommend the use of technical analysis as a sole means of investment research.
© 2023 Charles Schwab & Co., Inc. All rights reserved. Member SIPC. (1023-373U) ADP121824-00

[Schwab.com/Trading](https://www.schwab.com/trading)



Traders take many paths to reach their destination.
Can your platform get you there?

NINJATRADER ECOSYSTEM

Search apps and services to personalize the
NinjaTrader platform to meet your requirements.
Indicators, automated strategies, free tools & more.

Explore now at ninjatraderecosystem.com

Futures, foreign currency and options trading contains substantial risk and is not for every investor. Only risk capital should be used for trading and only those with sufficient risk capital should consider trading.

Cyclic personalities

by John Ehlers

Are there basic differences in the cyclic behavior of various commodities? Although theory predicts short-term cycles will come and go, there is no theory predicting different cyclic "personalities" for different commodities. My experience has been that cyclic trading is successful for some commodities but not for others. My curiosity about the cyclic behavior of different commodities was aroused during the development of a Maximum Entry Spectral Analysis program (MESA) that provided backtesting. Using the program, I measured the cycles of 12 commodities over the past two years. I found that each commodity does have its own cyclic personality. Moreover, commodities within the same category (meats, grains, metals) do not necessarily have similar cyclic personalities.

The measured histograms are shown in Figures 2-13. In speaking of cycle personalities, it is interesting to note that the S&P 500 is "schizophrenic," given its dual personality of 9 and 14 days.

Cycle measurements

I used perpetual daily contracts containing 500 records, and the average of the high and low for each day determined the price from which the cycle was measured. The program then measured the short-term cycles for each 50-day period, starting with record number 50. (The program needed the first 50 records to make the first measurement.) The dominant cycle and cyclic content were measured daily for records 50-500, a total of 451 measurements.

"Cyclic content" is a signal-to-noise ratio over 6 decibels. Noise is all of the non-cyclic energy with periods of less than 5 days and greater than 40 days. (Noise by this definition includes trendlines as well as the random day-to-day fluctuations of price.) My experience is that a cyclic content of at least 6 decibels is useful for trading. At lower values, the noise greatly influences any predictions based on cycles, reducing the accuracy of the predictions. Therefore, I discarded measurements having a cyclic content of less than 6 decibels.

The program created a histogram of those cycles with cyclic contents exceeding the 6-decibel threshold. The histogram is a cumulative count of how many times a 5-day cycle occurred, how many times a 6-day cycle occurred, and so on.

The result of this cumulative counting gives us two parameters with which to compare the cyclic behavior of the commodities. The first parameter is how "cyclic" was the overall contract—that is, how many times the threshold exceeded 6 decibels out of the 451 possibilities.

The second parameter is the dominant cycle—the contract's "personality." The dominant cycle is the histogram's peak response (or near-the-peak response in the case of asymmetry). I allowed a margin of ± 2 days surrounding the peak to account for error in computing the percentage of days that the contract exhibits its dominant cycle.

Dominant cycle

Figure 1 summarizes the measured results. The most cyclic commodities have dominant cycles that exceed the 6-decibel threshold 30%-40% of the time. In general, the most cyclic contracts also have the highest percentage of days exhibiting cyclic behavior—their "personalities" are most pronounced.

The measured histograms are shown in Figures 2-13. In speaking of cycle personalities, it is interesting to note that the S&P 500 is "schizophrenic," given its dual personality of 9 and 14 days.

After this experiment, I better understand why I could make cyclic trades in some commodities and not in others. I now focus my trading activity on those contracts that exhibit their dominant cycle the highest percentage of the time.

John F. Ehlers, Box 1801, Goleta, CA 93116, (805) 962-9477, is an electrical engineer working in electronic research and development, and has been a private trader for 10 years. He is a pioneer in introducing maximum entropy spectrum analysis to trading analysis through his MESA computer program.

Measured cyclic personalities

	Dominant cycle (days)	% Days within ±2 days of dominant cycle	% Total above 6 DB threshold
S&P 500	9 and 14	16.6 11.3	37.5
T. Bonds	9	16.0	36.1
Pork bellies	16	14.9	41.0
DM	13	14.9	31.7
Jap. Yen	11	10.0	20.4
Cattle	14	9.3	30.8
Copper	18	8.6	33.0
Cocoa	12	7.5	21.7
Wheat	7,11and 19	7.3	26.2
Sugar	7and 12	7.1	23.7
Gold	12	6.7	19.1
Hogs	19	6.2	29.9

FIGURE 1:

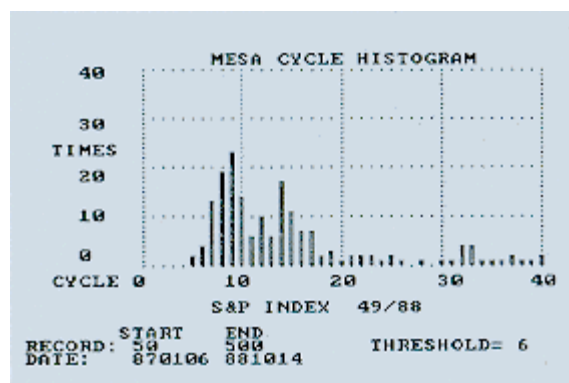


FIGURE 2:

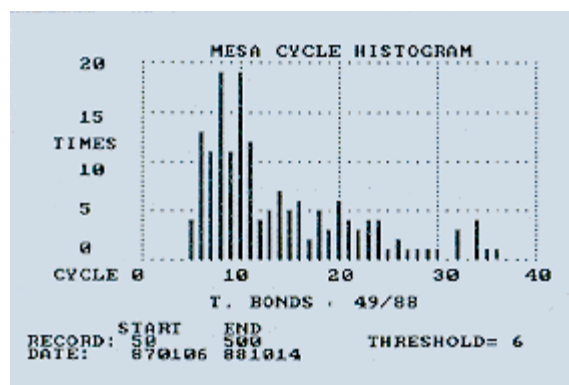


FIGURE 3:

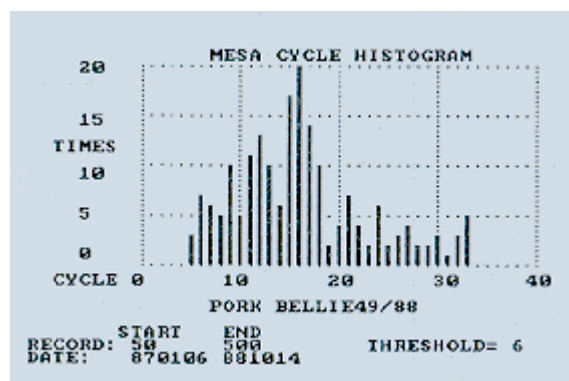


FIGURE 4:

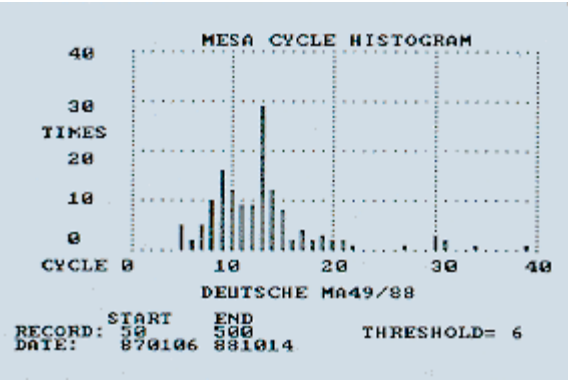


FIGURE 5:

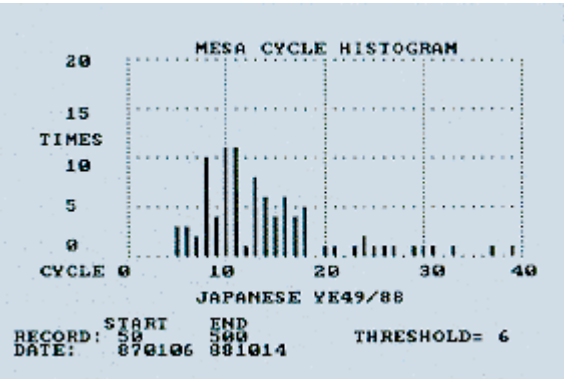


FIGURE 6:

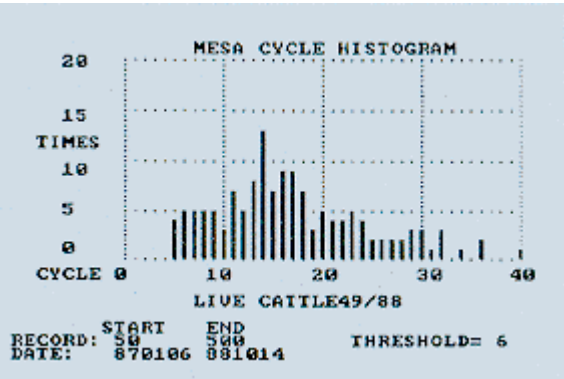


FIGURE 7:

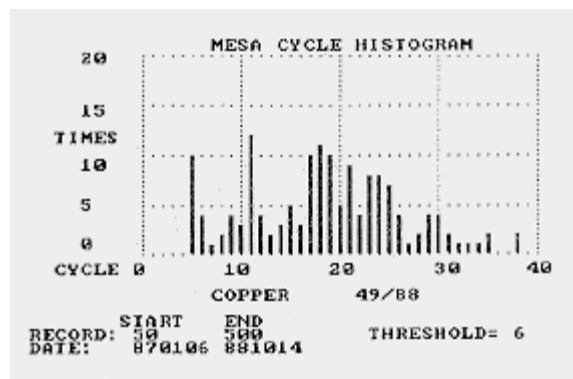


FIGURE 8:

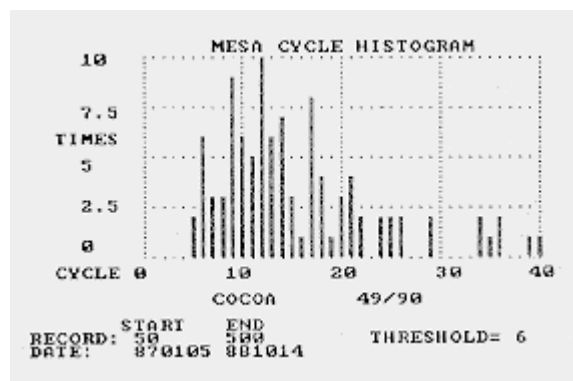


FIGURE 9:

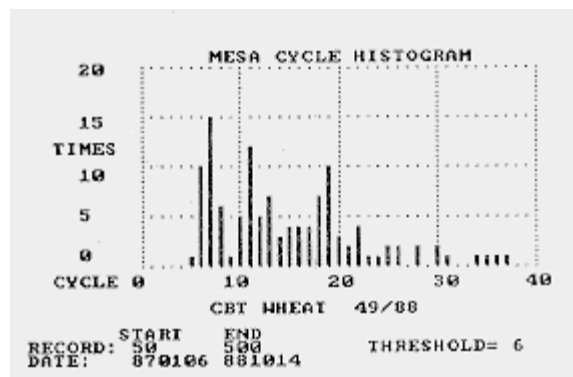


FIGURE 10:

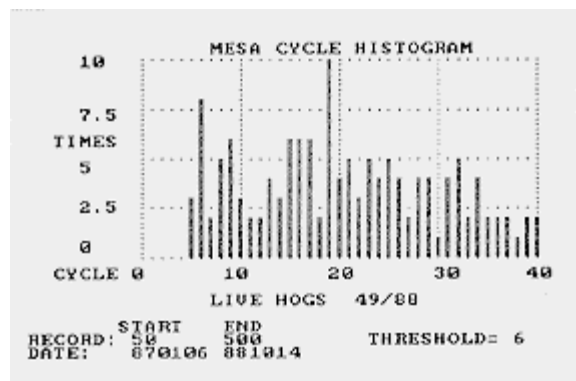


FIGURE 11:

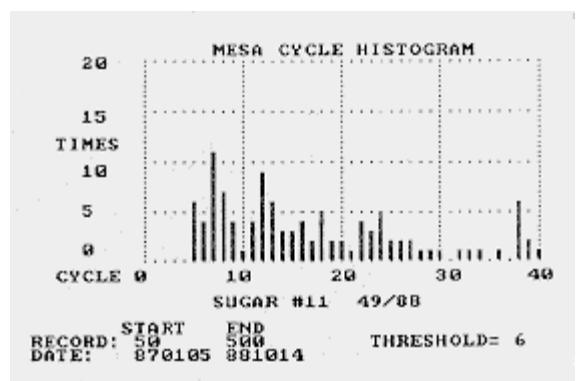


FIGURE 12:

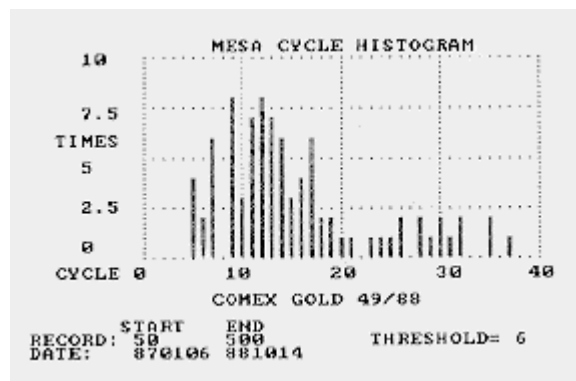


FIGURE 13:



Find the Best Bonds¹

Search our vast universe of
bonds with the free Bond
Search Tool to find the best
bond prices with no mark-ups

Rated Best
Online Broker 2023
for Bonds
by Benzinga¹



Interactive Brokers
Rated #1 ... Again
Best Online Broker
2023 by Barron's²

The best-informed
investors choose Interactive Brokers



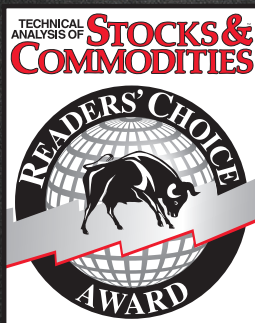
ibkr.com/find-bonds



Member - NYSE, FINRA, SIPC – Supporting documentation for any claims and statistical information will be provided upon request. [1] Rated best online broker for bond brokers according to Benzinga - Best Online Brokers for Bonds, February 22, 2023. [2] Interactive Brokers rated #1, Best Online Broker according to Barron's Best Online Brokers Survey of 2023: June 9, 2023. For more information see, ibkr.com/info - Barron's is a registered trademark of Dow Jones & Co. Inc.

07-IB23-1621CH1599

Can you imagine winning the
Stocks & Commodities Readers' Choice
Award for 30 consecutive years?



*Best Standalone
Analytical Software
in it's price category
1993-2023*



We can.

Get your free trial at MetaStock.com/TASC

This is neither a solicitation to buy or sell any type of financial instruments, nor intended as investment recommendations. All investment trading involves multiple substantial risks of monetary loss. Don't trade with money you can't afford to lose. Trading is not suitable for everyone. Past performance, whether indicated by actual or hypothetical results or testimonials are no guarantee of future performance or success. NO REPRESENTATION IS BEING MADE THAT ANY ACCOUNT WILL OR IS LIKELY TO ACHIEVE PROFITS OR LOSSES SIMILAR TO THOSE SHOWN. IN FACT, THERE ARE FREQUENTLY SHARP DIFFERENCES BETWEEN HYPOTHETICAL PERFORMANCE RESULTS OR TESTIMONIALS AND THE ACTUAL RESULTS SUBSEQUENTLY ACHIEVED BY ANY PARTICULAR TRADING PROGRAM. Furthermore, all internal and external computer and software systems are not fail-safe. Have contingency plans in place for such occasions. MetaStock assumes no responsibility for errors, inaccuracies, or omissions in these materials, nor shall it be liable for any special, indirect, incidental, or consequential damages, including without limitation losses, lost revenue, or lost profits, that may result from reliance upon the information presented.

SUBSCRIBE OR RENEW TODAY!

Every Stocks & Commodities subscription (regular and digital) includes:

- Full access to our Digital Edition
The complete magazine as a PDF you can download.
- Full access to our Digital Archives
That's 35 years' worth of content!
- Complete access to WorkingMoney.com
The information you need to invest smartly and successfully.
- Access to Traders.com Advantage
Insights, tips and techniques that can help you trade smarter.

1 year **\$89⁹⁹**

2 years **\$149⁹⁹**

3 years **\$199⁹⁹**

PROFESSIONAL TRADERS' STARTER KIT

A 5-year subscription to S&C magazine that includes everything above PLUS a free* book, *Charting The Stock Market: The Wyckoff Method*, all for a price that saves you \$150 off the year-by-year price! *Shipping & handling charges apply for foreign orders.

5 years **\$299⁹⁹**

That's around \$5 a month!



Visit **www.Traders.com** to find out more!